50 DATA ANALYST INTERVIEW Q&A

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SQL QUESTIONS (1-5)

1. What is the difference between WHERE and HAVING?

WHERE filters rows before grouping; HAVING filters groups after GROUP BY aggregation.

2. Explain JOIN types (INNER, LEFT, RIGHT, FULL)

INNER: matching rows only. LEFT: all left + matches. RIGHT: all right + matches. FULL: all rows from both.

3. What is a Primary Key vs Foreign Key?

Primary Key: Unique identifier for table rows. Foreign Key: References Primary Key in another table.

4. How do you find duplicate records?

Use GROUP BY with COUNT() > 1. Example: SELECT col, COUNT(*) FROM table GROUP BY col HAVING COUNT(*) > 1

5. What is the difference between UNION and UNION ALL?

UNION removes duplicates; UNION ALL keeps all rows including duplicates (faster).

SQL QUESTIONS (6-10)

6. What is a subquery?

A query nested inside another query. Used in SELECT, FROM, WHERE clauses to retrieve intermediate results.

7. Explain RANK() vs DENSE_RANK() vs ROW_NUMBER()

RANK: gaps in ranking. DENSE_RANK: no gaps. ROW_NUMBER: unique sequential numbers.

8. What is an INDEX and why use it?

Database structure to speed up data retrieval. Trade-off: faster reads, slower writes.

9. How to get 2nd highest salary?

SELECT MAX(salary) FROM employees WHERE salary < (SELECT MAX(salary) FROM employees)

10. What is normalization?

Organizing data to reduce redundancy. 1NF, 2NF, 3NF eliminate duplicate data and ensure data integrity.

PYTHON QUESTIONS (11-15)

11. What is Pandas and why use it?

Python library for data manipulation. Offers DataFrames for easy data cleaning, analysis, and transformation.

12. Explain DataFrame vs Series

Series: 1D labeled array. DataFrame: 2D table with rows and columns (like Excel sheet).

13. How to handle missing values?

dropna() to remove, fillna() to fill with value/mean/median, or interpolate() for time series.

14. What is GroupBy in Pandas?

Groups data by column(s) and applies aggregate functions like sum(), mean(), count().

15. List vs Tuple vs Dictionary?

List: mutable ordered. Tuple: immutable ordered. Dictionary: mutable key-value pairs.

PYTHON QUESTIONS (16-20)

16. What is NumPy and its use?

Numerical Python library for arrays, mathematical operations, and linear algebra. Faster than lists.

17. Merge vs Join vs Concat in Pandas?

Merge: SQL-like joins. Join: join on index. Concat: stack DataFrames vertically/horizontally.

18. How to read CSV/Excel files?

pd.read_csv('file.csv') and pd.read_excel('file.xlsx'). Export: df.to_csv() / df.to_excel()

19. What is Lambda function?

Anonymous one-line function. Example: lambda x: x*2. Used with apply(), map(), filter().

20. Explain apply() vs map()?

apply(): works on DataFrame/Series, any function. map(): Series only, element-wise transformation.

STATISTICS QUESTIONS (21-25)

21. Mean vs Median vs Mode?

Mean: average. Median: middle value (better for outliers). Mode: most frequent value.

22. What is Standard Deviation?

Measures data spread from mean. Low SD = data clustered; High SD = data scattered.

23. Explain p-value

Probability of results occurring by chance. $p < 0.05$ typically means statistically significant.

24. What is correlation?

Measures relationship strength between variables. Range: -1 to +1. 0 = no correlation.

25. Type I vs Type II error?

Type I: False positive (reject true null). Type II: False negative (accept false null).

STATISTICS QUESTIONS (26-30)

26. What is Normal Distribution?

Bell-shaped curve where data clusters around mean. 68-95-99.7 rule for standard deviations.

27. Explain Central Limit Theorem

Sample means approach normal distribution as sample size increases, regardless of population distribution.

28. What is hypothesis testing?

Statistical method to test assumptions. Compare p-value to significance level (α) to accept/reject hypothesis.

29. Variance vs Covariance?

Variance: spread of one variable. Covariance: how two variables change together (positive/negative).

30. What are outliers and how to handle?

Extreme values. Handle: remove, transform (log), or use robust methods (median instead of mean).

EXCEL & POWERBI (31-35)

31. VLOOKUP vs XLOOKUP vs INDEX MATCH?

VLOOKUP: lookup right. XLOOKUP: any direction (newer). INDEX MATCH: most flexible, any direction.

32. What is Pivot Table?

Excel tool to summarize, analyze, and reorganize data. Drag-drop fields to create dynamic reports.

33. Explain DAX in PowerBI

Data Analysis Expressions. Formula language for calculations in PowerBI (like Excel formulas).

34. CALCULATE function in DAX?

Most powerful DAX function. Modifies filter context to perform calculations across different dimensions.

35. What is data modeling in PowerBI?

Creating relationships between tables (star/snowflake schema) for efficient analysis and reporting.

DATA CLEANING & ANALYSIS (36-40)

36. How to handle missing data?

Remove rows (dropna), fill with mean/median/mode, forward/backward fill, or use ML imputation.

37. What is data normalization?

Scaling features to same range (0-1 or standardized). Prevents bias toward large-scale features.

38. Explain ETL process

Extract: get data from sources. Transform: clean and structure. Load: insert into database/warehouse.

39. How to detect data quality issues?

Check for nulls, duplicates, outliers, inconsistent formats, invalid values, and incorrect data types.

40. What is A/B testing?

Comparing two versions (A vs B) to determine which performs better. Statistical test on control vs treatment.

BEHAVIORAL QUESTIONS (41-45)

41. Tell me about a data project you're proud of

Use STAR method: Situation, Task, Action, Result. Focus on impact (revenue, efficiency, insights).

42. How do you handle tight deadlines?

Prioritize tasks, break into smaller chunks, communicate early if delays expected, focus on MVP first.

43. Describe a time you made a mistake

Own it, explain lesson learned, show how you prevented recurrence. Focus on growth mindset.

44. How do you explain technical findings to non-technical stakeholders?

Use visualizations, analogies, avoid jargon, focus on business impact, tell a story with data.

45. Why do you want to be a Data Analyst?

Passion for insights, problem-solving, impact on decisions, continuous learning. Be genuine and specific.

BONUS QUESTIONS (46-50)

46. What is the difference between data warehouse and database?

Database: transactional (OLTP), current data. Warehouse: analytical (OLAP), historical data, optimized for queries.

47. Explain regression vs classification

Regression: predict continuous values (price, temperature). Classification: predict categories (yes/no, spam/not spam).

48. What are KPIs? Give examples.

Key Performance Indicators. Metrics tracking goals. Examples: Conversion rate, CAC, churn rate, ROI, NPS.

49. How do you stay updated with data analytics trends?

Follow blogs, take courses, attend webinars, practice on Kaggle, read research papers, join communities.

50. What tools are you proficient in?

Tailor to job: SQL, Python (Pandas, NumPy), Excel, PowerBI/Tableau, Git, Jupyter. Mention specific projects.

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